Zuu Crew
Machine Learning Academy

MINI PROJECT 01

Designing a Modern Data Platform for ABC Hub

Scenario

You have recently joined ABC Hub, a rapidly growing entertainment company that provides both digital streaming services and physical media rentals through a single online platform. Customers can subscribe to monthly streaming plans to watch movies, TV series, documentaries, and music through the company's website and mobile application. Customers can also rent physical copies of movies, books, music albums, and video games, which are delivered to their homes and returned after the rental period.

The platform allows customers to:

- Register and manage their accounts.
- Subscribe to streaming plans.
- Browse and search digital and physical content.
- Stream digital content online.
- Rent physical items.
- Add unavailable items to a wish list.
- Make online payments.
- Rate and review content.
- Receive personalised recommendations.

To support these services, the company operates several operational systems, including:

- Customer Management
- Content Management
- Rental Management
- Subscription Management
- Inventory Management
- Payment Processing
- Delivery Management
- Customer Support

Although these systems support daily business operations, management has identified several challenges:

- Business reports take a long time to prepare.
- Different departments often report different business figures.
- Historical business trends are difficult to analyse.
- Customer behaviour across streaming and physical rentals cannot be analysed together.

- Business users rely heavily on IT teams to prepare reports.

The company plans to expand internationally over the next five years and expects significant growth in both customers and data volume.

As a newly appointed Data Engineer, you have been asked to analyse the business requirements, design a modern data platform, and implement an ETL solution that supports reporting, business intelligence, and future business growth.

Questions

Question 1 (10 Marks)

Analyse the business scenario and identify the major operational systems that generate data within the organisation.

Draw a high-level diagram illustrating how these operational systems support the business and interact with one another.

Question 2 (10 Marks)

Management has identified several reporting challenges.

Analyse these challenges and recommend a suitable solution to improve reporting and analytical capabilities.

Support your answer with an appropriate high-level architecture diagram.

Question 3 (15 Marks)

Gathering Analytical Requirements

Following the initial meeting, management and the Data Science team have provided the following high-level analytical requirements.

Management would like to analyse:

- Customer activity
- Streaming performance
- Physical rental trends
- Revenue
- Content popularity
- Inventory utilisation

The Data Science team has also requested access to historical customer activity and content usage data to support future machine learning and recommendation models.

As the Data Engineer, you believe these requirements are too high-level to begin designing the analytical platform.

Your task is to gather the missing business requirements.

Identify the questions you would ask the stakeholders before designing the data platform.

Your questions should help clarify areas such as:

- Business objectives

- KPIs and business measures
- Required dimensions
- Time granularity
- Reporting frequency
- Historical data requirements
- Data quality expectations
- Performance expectations
- Security and access requirements
- Future scalability

Based on your answers, make reasonable assumptions where necessary.

For each assumption, briefly explain why it is appropriate.

Finally, prepare a refined set of analytical requirements that will be used as the foundation for designing the analytical data platform.

Example

Initial Requirement

Management would like to analyse customer activity.

Questions you may ask

- Do you need daily, weekly, monthly or yearly analysis?
- Should customer activity be analysed by country, city or branch?
- Do you want to compare subscription plans?
- Should inactive customers be included?
- What KPIs are important (stream count, rentals, revenue, watch duration)?
- How much historical data should be retained?

Example Assumptions

- Reports will support daily and monthly analysis.
- Customer location will be analysed at country and city levels.
- Customer activity includes both streaming and physical rentals.
- Historical data will be retained indefinitely.

Deliverables

Your submission should include:

- A list of stakeholder questions.
- Your assumptions and justifications.
- A refined analytical requirements document that will be used in the remaining questions.

Question 4 (10 Marks)

The company must retain complete historical information and provide full traceability of data for auditing and regulatory purposes.

Recommend a suitable data modelling approach to satisfy these requirements.

Support your answer with:

- A justification for your chosen approach.

- A diagram illustrating the model.

Question 5 (10 Marks)

Business analysts have reported that preparing reports requires combining a large number of tables, making analysis slow and difficult.

Recommend an approach that simplifies data access for analysts while maintaining business value.

Discuss:

- How your proposed solution improves analytical performance.
- The advantages of the approach.
- Any potential limitations or trade-offs.

Question 6 (10 Marks)

Design an end-to-end modern data platform that transforms operational data into trusted analytical data for business decision-making.

Your architecture should illustrate the complete data flow from operational systems to business users.

Clearly explain the purpose of each major component in your architecture.

Question 7 (15 Marks)

Designing an Analytical Dimensional Model

The Chief Data Officer has identified several recurring business reports that cannot be efficiently produced from the operational systems.

Instead of analysing individual streaming sessions or rental transactions, management requires pre-calculated analytical datasets that summarise business performance over time.

Design suitable Fact and Dimension tables for the following analytical scenarios.

Scenario A – Customer Daily Activity

Design an analytical model that stores one record per customer per day.

Include suitable measures such as:

- Number of streaming sessions
- Total streaming duration
- Number of physical rentals
- Number of returned items
- Total amount spent
- Number of support tickets raised

Scenario B – Content Monthly Performance

Design an analytical model that stores one record per content item per month.

Include suitable measures such as:

- Total streams

- Total rental count
- Revenue generated
- Average customer rating
- Number of wish list additions

Scenario C – Inventory Utilisation

Design an analytical model that stores one record per physical inventory item per day.

Include suitable measures such as:

- Number of rentals
- Number of returns
- Number of days available
- Inventory utilisation percentage

For each analytical model, you should:

- Identify the grain.
- Design the Fact table(s).
- Design the required Dimension tables.
- Identify the primary keys and foreign keys.
- Identify the business measures and explain why they belong in the Fact table.
- Draw a Star Schema diagram.

> *Note: Your design should represent business-oriented analytical models, rather than simply copying the operational database structure.*

Question 8 (20 Marks)

Implementing an ETL Pipeline using Apache NiFi and PostgreSQL

ABC Hub has selected Apache NiFi as its ETL platform and PostgreSQL as its analytical database.

Using the operational database provided, implement a complete ETL pipeline in Apache NiFi that populates the analytical data model designed in Question 7.

Your implementation should demonstrate the complete ETL process from extracting operational data to loading the analytical database.

Task 1 – Data Extraction (4 Marks)

Develop a NiFi workflow to extract data from the operational PostgreSQL database.

Your pipeline should:

- Connect to the operational database.
- Extract data from the required operational tables.
- Read data incrementally where appropriate.
- Transfer the extracted data to the transformation stage.

Task 2 – Data Transformation (7 Marks)

Implement the necessary transformations using appropriate NiFi processors.

Your pipeline should:

- Validate mandatory fields.
- Handle null or invalid values.
- Remove or manage duplicate records.
- Apply the required business transformations.
- Generate the aggregated datasets required for the analytical models designed in Question 7.

Task 3 – Loading the Analytical Database (5 Marks)

Load the transformed data into the PostgreSQL analytical database.

Your implementation should:

- Load Dimension tables before Fact tables.
- Maintain referential integrity.
- Prevent duplicate analytical records during repeated executions.

Task 4 – Pipeline Management (4 Marks)

Configure the NiFi workflow for production execution.

Your implementation should include:

- Error handling using failure relationships.
- Logging of failed records.
- Appropriate processor scheduling.
- Clear processor naming and workflow organisation.

Submission Requirements

> *A Drive link for uploading your submission will be shared closer to the deadline.*

1. Report (PDF)

A report answering Questions 1–8, including:

- Business analysis
- Architecture diagrams
- Data modelling diagrams
- Star Schema diagrams
- Data platform architecture
- Explanations and justifications for all design decisions

2. Apache NiFi Flow

Submit one of the following:

- Exported NiFi Flow Definition (JSON), or
- Exported NiFi Template (if applicable)

The workflow should successfully implement the ETL pipeline described in Question 8.

3. PostgreSQL SQL Scripts

Include all SQL scripts required to recreate the analytical database, including:

- Database creation script

- Schema creation scripts
- Dimension table creation scripts
- Fact table creation scripts
- Supporting views, indexes, or stored procedures (if used)

4. Source Code and Supporting Files

Include any additional files required to execute your solution, such as:

- SQL transformation scripts
- Configuration files
- Sample datasets (if required)
- Supporting scripts used within the ETL pipeline

5. Screenshots

Provide screenshots demonstrating:

- The completed Apache NiFi workflow
- Successful execution of the ETL pipeline
- PostgreSQL analytical tables populated with data
- Sample query results from the analytical database

6. README File

Include a README file describing:

- Project structure
- Software versions used
- Installation steps
- Database configuration
- Instructions for executing the NiFi workflow
- Any assumptions made during implementation

Assessment Guidelines

- Clearly state any assumptions made during the assignment.
- All diagrams should be clearly labelled and easy to read.
- Follow data engineering best practices when designing your data platform and ETL pipeline.
- The Apache NiFi workflow should use meaningful processor names and be organised for readability and maintainability.
- The PostgreSQL analytical database should successfully support the analytical models designed in Question 7.
- Marks will be awarded for correctness, completeness, clarity of design, implementation quality, and adherence to best practices.
- Late submissions will be handled according to the module assessment policy.